Edgardo Villar-Sepúlveda

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Curriculum Vitae

Research profile

My research lies at the interface of applied mathematics, nonlinear dynamics, and mathematical biology, with a focus on pattern formation in nonlinear partial differential equations. I investigate Turing and Turing–wave instabilities, bifurcations, nonlinear waves, localised patterns, and homoclinic snaking in reaction–diffusion–transport systems. My recent research combines asymptotic analysis and dynamical systems, with a particular focus on Maxwell points and beyond-all-order asymptotics. Code accompanying several of my papers is available on my website.

Positions held

2026–Present **TUM Global Postdoc Fellow**, *School of Computation, Information and Technology*, Technical University of Munich, Germany.

2024–2026 **Teaching Associate**, *School of Engineering Mathematics and Technology*, University of Bristol, UK.

2021–2024 **Teaching Assistant**, *School of Computer Science, Electrical and Electronic Engineering and Engineering Mathematics*, University of Bristol, UK.

2020 **Engineering Teacher**, *Facultad de Ingeniería y Ciencias*, Universidad Adolfo Ibáñez, Viña del Mar, Chile.

2018–2020 **Engineering Teacher**, *Departamento de Matemática*, Universidad Técnica Federico Santa María, Valparaíso, Chile.

Education

2021–2025 **PhD in Engineering Mathematics**, *University of Bristol*, UK.
Supervisor: Prof. Alan Champneys. **Date of award:** 17 June 2025. **Thesis:** General approaches to pattern formation in reaction–diffusion systems.

2018–2019 **MSc in Mathematics**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile.
Supervisor: Dr. Pablo Aguirre Olea.

2014–2019 **BSc in Mathematical Engineering**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile, with Academic Distinction.

Teaching training

2026 **ProLehre Onboarding**, *Technical University of Munich (completed October 2026).*
Elective module: Supervising Students' Theses (16 work units), covering structured supervision meetings, constructive feedback, and criterion-based marking. Work units are recognised towards the advanced level of the Certificate for Teaching in Higher Education of the Bavarian Universities.

Fellowships and Awards

2026–Present **TUM Global Postdoc Fellowship**, *Technical University of Munich*, Host: Prof. Christian Kühn, ∼€75,000.
Awarded through a competitive international selection process to support independent postdoctoral research at TUM.

2025 **Patrick Woods Prize**, *University of Bristol*.
Best PhD Thesis of the School of Engineering Mathematics and Technology for the academic year 2024–2025. £2,500.

2024 **Student Travel Award**, *SIAM*.
Funded to attend the SIAM Conference on Nonlinear Waves and Coherent Structures in Baltimore, Maryland, USA. US$975.

2023 **Student Travel Award**, *SIAM*.
Funded to attend the SIAM Conference on Applications of Dynamical Systems in Portland, Oregon, USA. US$950.

2021 **Bristol PLUS Award**, *University of Bristol*.
Awarded to fewer than 5% of students annually for exceptional extracurricular engagement, including over 90 hours of teaching assistance and activities supporting professional and personal development.

2020 **Beca Chile de Doctorado en el Extranjero**, *Agencia Nacional de Investigación y Desarrollo*, Santiago, Chile.
Scholarship awarded through a competitive merit-based selection process to fund my PhD and living expenses. ∼US$200,000.

2019 **Academic Merit Award**, *Universidad Técnica Federico Santa María*, Valparaíso.
Awarded to the top student among all those who entered the university in 2014, across all programmes. ∼US$5,000.

2014–2019 **Dean's Honor List**, *Universidad Técnica Federico Santa María*, Valparaíso.
Ranked among the best students at my university each year. In 2018 and 2019, I ranked second among all enrolled students, across all entry-year cohorts.

2013 **Beca Puntaje Nacional**, *Universidad Técnica Federico Santa María*, Valparaíso.
Scholarship awarded to fund my undergraduate and master's studies for obtaining a perfect score in the University entrance test in Mathematics. ∼US$35,000.

Publications

Published Journal Articles

[13] **Edgardo Villar-Sepúlveda**. Beyond-all-order asymptotics for homoclinic snaking of localised patterns in reaction-transport systems. *Nonlinearity*, 39(9):095025, 2026.

[12] **Edgardo Villar-Sepúlveda**, Alan R Champneys, Davide Cusseddu, and Anotida Madzvamuse. Pattern formation of bulk-surface reaction-diffusion systems in a ball. *SIAM Journal on Applied Mathematics*, 86(1):21–51, 2026.

[11] Andrew L Krause, Václav Klika, **Edgardo Villar-Sepúlveda**, Alan R Champneys, and Eamonn A Gaffney. Pattern localization in the Swift–Hohenberg equation via slowly varying spatial heterogeneity. *SIAM Journal on Applied Dynamical Systems*, 24(4):2804–2847, 2025.

[10] Fasma Diele, Andrew L Krause, Deborah Lacitignola, Carmela Marangi, Angela Monti, and **Edgardo Villar-Sepúlveda**. Transient instability and patterns of reactivity in diffusive-chemotaxis soil carbon dynamics. *Bulletin of Mathematical Biology*, 87(11):162, 2025.

[9] **Edgardo Villar-Sepúlveda**, Alan R Champneys, and Andrew L Krause. Designing reaction-cross-diffusion systems with Turing and wave instabilities. *Journal of Mathematical Biology*, 91(4):37, 2025.

[8] Molly Brennan, Andrew L Krause, **Edgardo Villar-Sepúlveda**, and Christopher B Prior. Pattern formation as a resilience mechanism in cancer immunotherapy. *Bulletin of Mathematical Biology*, 87(8):106, 2025.

[7] Chunyi Gai, **Edgardo Villar-Sepúlveda**, Alan Champneys, and Michael J Ward. An asymptotic analysis of spike self-replication and spike nucleation of reaction-diffusion patterns on growing 1-D domains. *Bulletin of Mathematical Biology*, 87(4):48, 2025.

[6] **Edgardo Villar-Sepúlveda** and Alan Champneys. Amplitude equations for wave bifurcations in reaction–diffusion systems. *Nonlinearity*, 37(8):085012, 2024.

[5] Alfredo E Calderón and **Edgardo Villar-Sepúlveda**. Piecewise contracting maps on the interval: Hausdorff dimension, entropy, and attractors. *Canadian Mathematical Bulletin*, 67(2):318–327, 2024.

[4] **Edgardo Villar-Sepúlveda** and Alan Champneys. Computation of Turing bifurcation normal form for n-component reaction-diffusion systems. *ACM Transactions on Mathematical Software*, 49(4):1–24, 2023.

[3] **Edgardo Villar-Sepúlveda** and Alan Champneys. Degenerate Turing bifurcation and the birth of localized patterns in activator-inhibitor systems. *SIAM Journal on Applied Dynamical Systems*, 22(3):1673–1709, 2023.

[2] **Edgardo Villar-Sepúlveda**, Pablo Aguirre, and Víctor F Breña-Medina. A case study of multiple wave solutions in a reaction-diffusion system using invariant manifolds and global bifurcations. *SIAM Journal on Applied Dynamical Systems*, 22(2):918–950, 2023.

[1] **Edgardo Villar-Sepúlveda** and Alan R Champneys. General conditions for Turing and wave instabilities in reaction-diffusion systems. *Journal of Mathematical Biology*, 86(3):39, 2023.

Preprints

[P1] Alfredo E Calderón and **Edgardo Villar-Sepúlveda**. Unifying topological entropy notions for piecewise continuous maps on the interval. *arXiv:2404.10867*, 2024.

Teaching

2024–2026 **Teaching Associate, University of Bristol**.

MSc level: I was Unit Director for the MSc course *Mathematical Modelling in Biology, Medicine and Public Health* (∼15 students), where I designed lectures and assessments, and coordinated the teaching team. I also taught students across different MSc programmes, including Data Science, Engineering Mathematics, and Financial Technology with Data Science. In 2025, I co-led half of the MSc course *Intro to AI and Data Analytics* (∼60 students) and designed the associated coursework. In addition, throughout this period, I taught *Visual Analytics* (∼250 students) and contributed to *Software Development: Programming and Algorithms* (∼40 students), *Statistical Computing and Empirical Methods*, and *Large-Scale Data Engineering for Business*.

Undergraduate level: I taught *Engineering Mathematics 1* (∼1,050 students), and contributed to *Engineering Mathematics 2* (∼850 students), *Data Engineering at Scale*, and *Data Science and Applied Statistics*. In addition, I assessed student project work in two courses: for *Introduction to Computer Programming*, I reviewed 17 undergraduate coding projects and questioned students about their code and implementation choices when they presented their work; for *Mathematical and Data Modelling*, I supervised 3 group projects proposed by industry partners and marked their reports. Also, for the *Discrete Mathematics* course, I developed three sets of supplementary materials based on textbook questions: one containing deliberately incorrect proofs, another containing corrected versions of those proofs, and a third providing complete, correct solutions. These resources helped students learn to identify and correct errors and construct rigorous proofs.

I also supervised Financial Technology with Data Science master's projects and served as a personal tutor for undergraduate and master's students, helping them adapt to different educational systems and assessment expectations.

2021–2024 **Teaching Assistant, University of Bristol**.

MSc level: Throughout this period, I was a Teaching Assistant for *Data Science Mini Project* (∼100 students), serving as Lead Teaching Assistant in 2022. In this MSc pre-dissertation course, I supervised group research projects proposed by industry partners and marked 10 project reports. In 2023, I co-led that course with an incoming lecturer, whom I supported throughout its delivery. I also assisted with the courses *Software Development: Programming and Algorithms* and *Visual Analytics*.

Undergraduate level: I assisted with *Data Science and Applied Statistics* (∼60 students), *Introduction to Data Science*, *Perspectives in Mathematics*, *Mathematical and Data Modelling*, *Introduction to Computer Programming*, *Nonlinear Dynamics and Chaos* (∼45 students), *Numerical Methods in MATLAB* (∼60 students), *Applied Linear Algebra*, *Engineering Mathematics 1*, *Engineering Mathematics 2*, and *Discrete Mathematics*. I also supported final-year BSc students with their dissertations through the *Final Year Project* course.

In addition, I contributed to the Advanced Mathematics Support Programme's *Problem Solving Matters* course, which aims to develop Year 12 students' mathematical thinking and problem-solving skills.

2020 **Engineering Teacher, Universidad Adolfo Ibáñez**.

Undergraduate level: I taught *Linear Algebra* and multivariable *Integral Calculus* (∼30 students).

2018–2020 **Engineering Teacher, Universidad Técnica Federico Santa María**.

Undergraduate level: I taught *Algebra and Geometry* (∼40 students), single- and multivariable *Differential Calculus* (∼40 students), and single- and multivariable *Integral Calculus* (∼40 students). Across my Engineering Teacher positions at Universidad Adolfo Ibáñez and Universidad Técnica Federico Santa María, I also developed LaTeX-based teaching and assessment materials, including collections of examination questions, additional exercises, and worked solutions. Additionally, I organised voluntary weekly problem-solving sessions, sharing a selection of problems with students in advance so that they could attempt them independently. At these meetings, we worked through the problems collectively, with me providing detailed explanations and addressing students' questions and difficulties.

2014–2017 **Teaching Assistant, Universidad Técnica Federico Santa María**.

Undergraduate level: I assisted with *Differential Geometry of Curves and Surfaces*, single- and multivariable *Differential Calculus* (∼80 students), single- and multivariable *Integral Calculus* (∼40 students), and *Linear Algebra* (∼40 students).

MSc thesis supervision

Technical University of Munich: Mathematics

2026–Present **Barış Arslan**, *Exploratory numerical continuation of reaction–diffusion systems.* Co-supervised with Christian Kühn.

University of Bristol: Financial Technology with Data Science

2025 **Tianyu Dong**, *Advanced modeling methods for stock market: time series prediction and multi-market exploratory clustering evaluation.*

2025 **Hanwen Li**, *Advanced methods to forecast the stock market: A unified benchmark of LSTM, CNN–LSTM and Attention–LSTM for long-horizon stock prediction.*

2025 **Dushyant Goel**, *Predictive modeling and risk analysis for UK start-up fundraising markets with an interactive decision-support tool for investors and founders.*

2025 **Biao Yin**, *Deep learning approaches for forecasting the S&P 500 Index: A comparative study of LSTM, CNN, and a hybrid model.*

2025 **Zhengbo Xu**, *UK tax incentives and startup financing dynamics: A multi-platform study of the early-stage ecosystem.*

2025 **Tangrong Li**, *Time-series forecasting of stock prices using transformer variants: A critical evaluation of Informer, Autoformer, and PatchTST.*

2025 **Li Yin**, *Research on the adaptability of advanced deep learning methods in stock prediction of different companies.*

University of Bristol: Engineering Mathematics

2022 **Weiyi Yuan**, *Mathematical modelling of infection spread within a relatively static crowd*.
Co-supervised with Alan Champneys.

Conferences, Seminars, and Study Groups

Invited Talks

Jul 2026 **Dynamics Days Europe**, *Rectorate of Universidade NOVA de Lisboa and Colégio Almada Negreiros*, Lisbon, Portugal, Minisymposium: Singularly perturbed multiscale systems.
Maxwell points, the origin of nontrivial connections in multiscale systems

May 2026 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *Concordia University*, Montréal, Canada, Minisymposium: Coherent structures in biology and social sciences.
Maxwell points: The origin of homoclinic snaking and localised patterns beyond all asymptotic orders

May 2026 **Applied Nonlinear Dynamics seminar**, *University of Leeds*, UK.
Maxwell points, the origin of nontrivial connections in multiscale systems

May 2026 **Oberseminar Dynamics seminar**, *Technical University of Munich*, Germany.
Maxwell points, the origin of nontrivial connections in multiscale systems

Mar 2026 **British Applied Mathematics Colloquium**, *University of East Anglia*, Norwich, UK, Minisymposium: The next generation: frontiers in nonlinear waves and pattern formation.
Wave instabilities: the second face of Turing

May 2025 **SIAM Conference on Applications of Dynamical Systems**, *Sheraton Denver Downtown Hotel*, Denver, Colorado, US, Minisymposium: Waves in continuous and discrete domains: theory and applications.
Exponential asymptotics for the study of 'localised spatiotemporal patterns' on the line

May 2025 **Dynamical Systems seminar**, *University of Hamburg, Germany*.
Homoclinic snaking & localised patterns beyond all asymptotic orders

Feb 2025 **Dynamical Systems seminar**, *Leiden University, The Netherlands*.
Homoclinic snaking & localised patterns beyond all asymptotic orders

Jun 2024 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *Lord Baltimore Hotel*, Baltimore, Maryland, US, Minisymposium: Reaction–diffusion mechanisms for localised patterns and waves.
Travelling and standing waves as a consequence of cross-diffusion

May 2023 **SIAM Conference on Applications of Dynamical Systems**, *DoubleTree by Hilton Hotel*, Portland, Oregon, US, Minisymposium: Pattern formation in nature: from Busse balloons to homoclinic snaking.
Spatial instabilities as a consequence of Turing or transcritical bifurcations in bulk–surface reaction–diffusion equations on a sphere

Apr 2023 **British Applied Mathematics Colloquium**, *University of the West of England*, Bristol, UK, Minisymposium: Dynamics of reaction–transport systems.
Homoclinic snaking & localised patterns beyond all asymptotic orders

Nov 2022 **Mathematics of beyond all-orders phenomena**, *Isaac Newton Institute for Mathematical Sciences*, Cambridge, UK, Workshop.
Homoclinic snaking and the emergence of localised patterns beyond all asymptotic orders

Sep 2022 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *University of Bremen*, Germany, Minisymposium: Pattern formation, cell polarity and morphogenesis.
Turing patterns in general n-component reaction–diffusion equations posed on different geometries

Organisation of Minisymposia and Seminars

May 2026 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *Concordia University*, Montréal, Canada.
Organised Minisymposium (separate from my invited talk above): *Pattern formation, homoclinic snaking and modulation equations in non-standard systems* with Sebastian Suckau and Lütfiye Masur

2022–2024 **Engineering Mathematics Research Group seminars**, *University of Bristol*.
I organised seminars for weekly meetings we held with the members of the Engineering Mathematics Research Group

Contributed Talks

Jun 2025 **British Mathematical Colloquium & British Applied Mathematics Colloquium**, *University of Exeter*, UK.
Homoclinic snaking & localised patterns beyond all asymptotic orders

Feb 2025 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada, Online.
Exponential asymptotics to study the extent to which localised patterns exist

Apr 2024 **British Applied Mathematics Colloquium**, *University of Newcastle*, UK.
Turing and wave instabilities in reaction–diffusion systems with cross-diffusion

Nov 2023 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada, Online.
Diffusion-driven instabilities in reaction–diffusion systems with cross-diffusion

May 2023 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada.
Spatial instabilities as a consequence of Turing or transcritical bifurcations in bulk–surface reaction–diffusion equations on a sphere

Jan 2023 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada, Online.
Spatial and spatiotemporal oscillations in reaction–diffusion equations

Jun 2022 **Early Career Researchers in Mathematics**, *University College London*, London, UK.
Turing patterns in bulk–surface reaction–diffusion equations

Apr 2022 **British Applied Mathematics Colloquium**, *University of Loughborough*, UK, Online.
Turing instabilities in bulk–surface reaction–diffusion equations on a sphere

Sep 2021 **Celebrating Alan Turing's work in mathematical biology**, *University of Sheffield*, UK.
Necessary and sufficient conditions for an $n-$component system to show a Turing or Turing-Hopf instability pattern

Jan 2020 **World Conference on Natural Resource Modeling**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile.
Spatial diffusion in a predator-prey model with a strong Allee effect on the prey and a ratio-dependent functional response.

Apr 2019 **XXXII Jornada Matemática de la Zona Sur**, *Universidad de Magallanes*, Punta Arenas, Chile.
Ondas viajeras y caos en un modelo depredador-presa con respuesta funcional razón-dependiente y difusión espacial.

Jan 2019 **Tercer Workshop de Modelamiento Matemático de Sistemas Biológicos**, *Universidad Tecnológica Metropolitana*, Santiago, Chile.
Análisis de la influencia de la difusión espacial en un modelo depredador-presa.

Internal talks

May 2026 **Engineering Mathematics seminar**, *University of Bristol*.
Maxwell points, the origin of nontrivial connections in multiscale systems

Apr 2025 **Engineering Mathematics Postgraduate seminar**, *University of Bristol.*
Homoclinic snaking and localised patterns beyond all asymptotic orders

Dec 2022 **Engineering Mathematics seminar**, *University of Bristol.*
Homoclinic snaking & localised patterns beyond all asymptotic orders

Jun 2022 **PhD Away Day (with fellow Bristol PhD students)**, *University of Oxford*, Prize for the exceptional scientific content of my talk.
Pattern formation in reaction–diffusion equations

Dec 2021 **Engineering Mathematics Postgraduate seminar**, *University of Bristol.*
Necessary and sufficient conditions for a reaction–diffusion equation with n components to show a Turing or Turing-Hopf instability pattern

Study Groups

May 2024 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: The future of the high street.

Jan 2024 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Food security.

Nov 2023 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Maths for justice.

Oct 2022 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Recovery from the pandemic: Hospitality & leisure.

Jul 2022 **European Study Group with Industry**, *University of Kent*, Canterbury.
Theme: Alternate transforms for signal processing.

Jul 2021 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Covid safety in large events.

Academic and Administrative Service

Journal reviewing *Proceedings of the Royal Society A*; *Nonlinearity*; *Chaos, Solitons and Fractals*; *Physica D: Nonlinear Phenomena*; *IMA Journal of Applied Mathematics*; *Calculus of Variations and Partial Differential Equations*; *Journal of Nonlinear Science*; *Journal of Mathematical Biology*; *Computational and Applied Mathematics*; *Qualitative Theory of Dynamical Systems*; *Computational Methods for Differential Equations*; *Nonlinear Dynamics*; *Cognitive Computation*.

2025 **Provided academic references for students applying for PhD positions**.

2025 **Second marker for three MSc dissertations in Financial Technology with Data Science**.